

Spatial Gap Analysis

GBIF Gap Analysis - Geographic Coverage Assessment

gbifgaps

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Contents

Overview	2
1. Spatial Coverage Summary	2
1.1 Grid Resolution Comparison	2
1.2 Occurrence Distribution	3
2. Geographic Distribution Maps	4
2.1 Overall Coverage Map (10km)	4
2.2 Coverage Categories Map	5
2.3 Species Richness Map	6
3. Coverage by Basis of Record	7
3.1 Zero Coverage Rates	7
3.2 Coverage Comparison Across Basis Types	8
4. Priority Areas	9
4.1 Zero Coverage Cells	9
4.2 Priority Cells Map	10
4.3 Low Coverage Analysis	10
5. Key Findings	11
Spatial Coverage Summary	11
Session Info	12

Overview

This report analyzes **spatial gaps** in the national biodiversity occurrence data, examining geographic coverage patterns across EEA reference grids at 10km and 50km resolutions.

Key questions addressed:

- What proportion of the study area has biodiversity occurrence data?
- Where are the geographic gaps in sampling coverage?
- How does coverage vary by evidence type (basis of record)?
- Which areas should be prioritized for future sampling efforts?

1. Spatial Coverage Summary

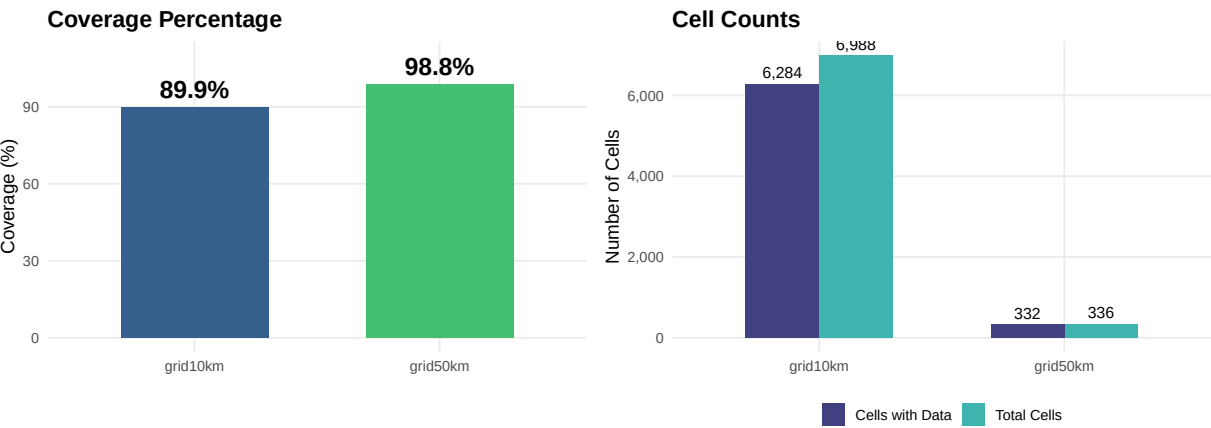
1.1 Grid Resolution Comparison

Table 1: Spatial Coverage by Grid Resolution

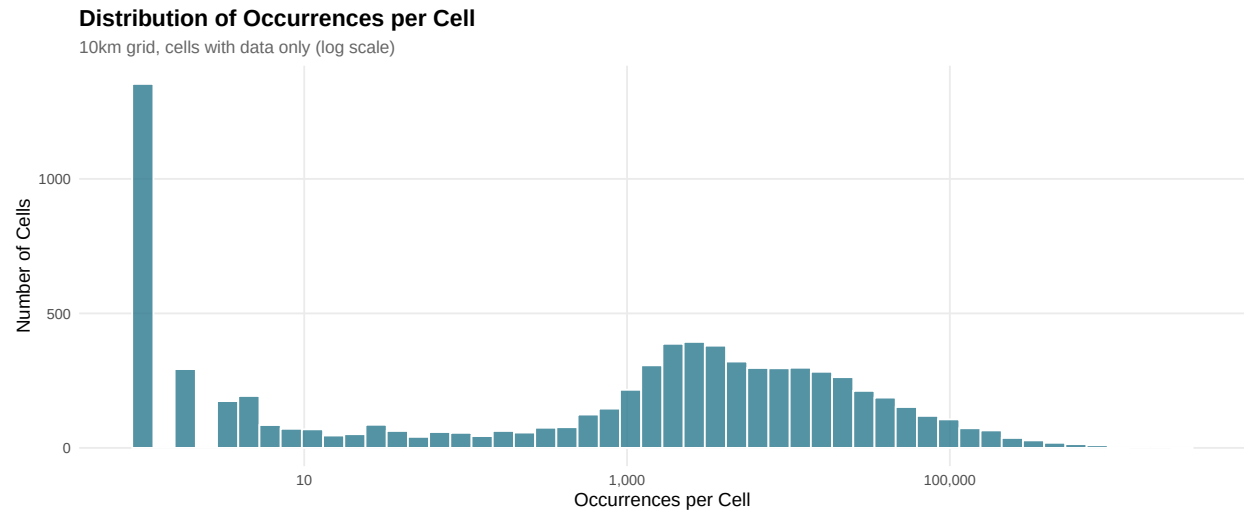
grid	n_cells_total	n_cells_with_data	cells_zero	pct_coverage	total_occurrences	mean_per_cell	median_per_cell
grid10km	6,988	6,284	704	89.9	134,235,995	19,209.5	2,354.5
grid50km	336	332	4	98.8	134,238,011	399,517.9	86,993.0

Spatial Coverage Overview

Comparing 10km and 50km EEA reference grids



1.2 Occurrence Distribution

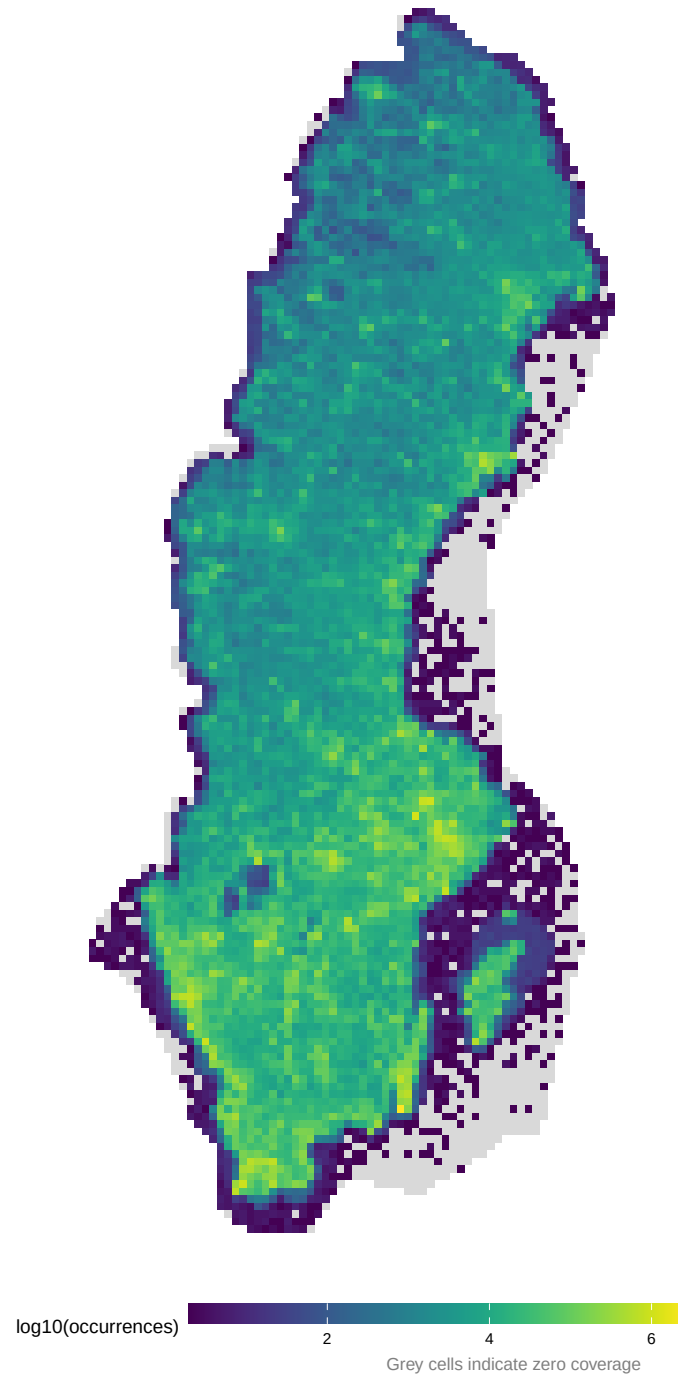


2. Geographic Distribution Maps

2.1 Overall Coverage Map (10km)

Geographic Distribution of Occurrence Data

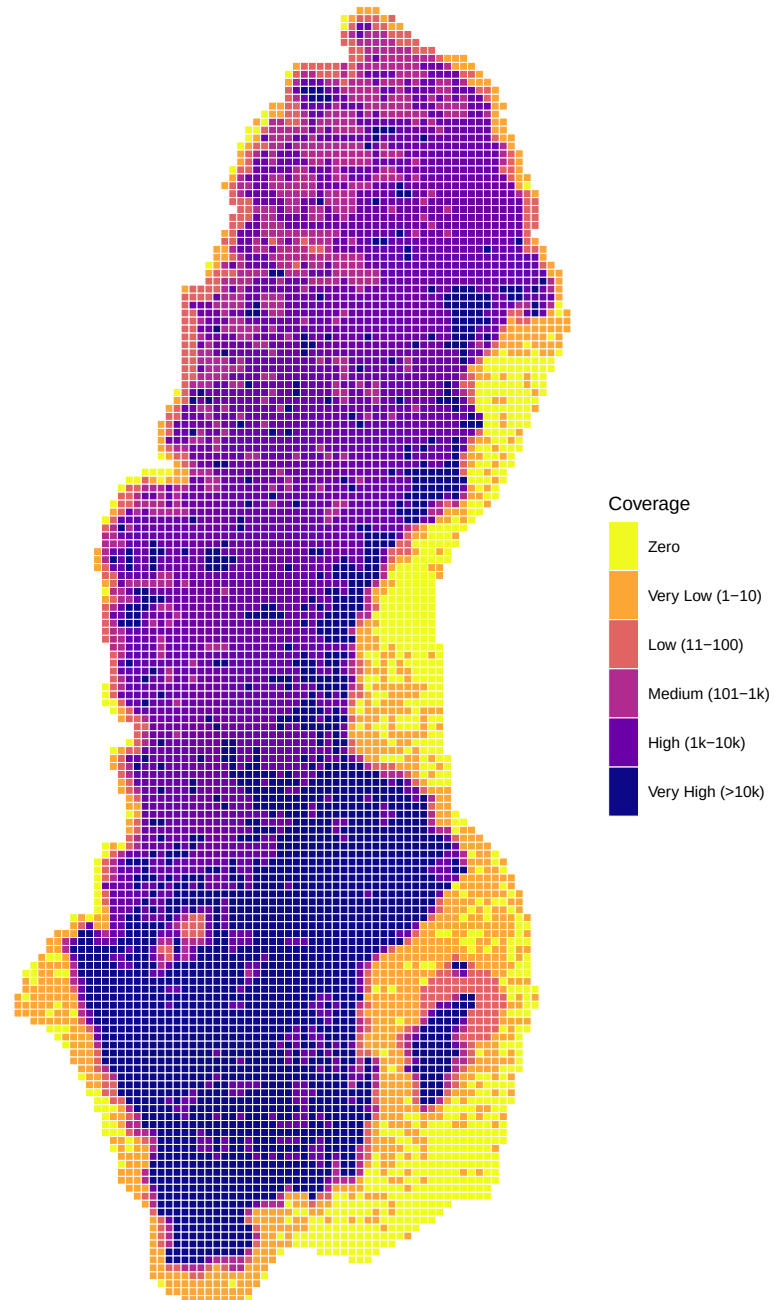
10km EEA Grid



2.2 Coverage Categories Map

Coverage Categories

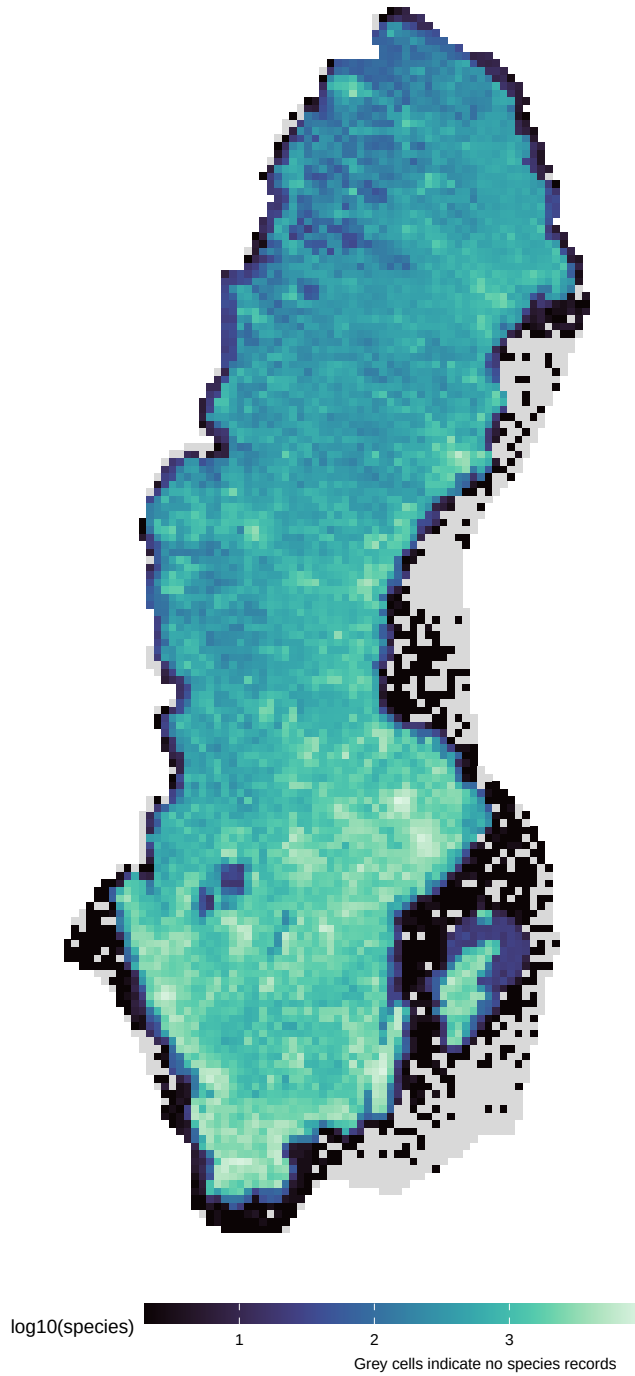
10km grid cells classified by occurrence count



2.3 Species Richness Map

Species Richness Distribution

Number of unique species per 10km cell

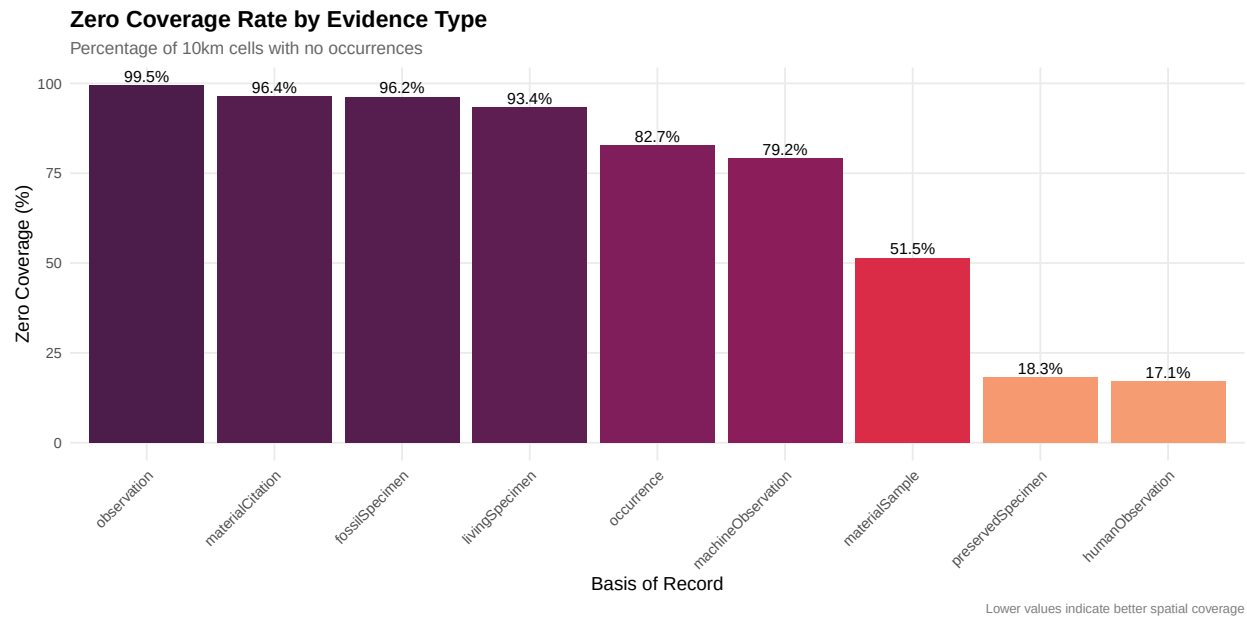


3. Coverage by Basis of Record

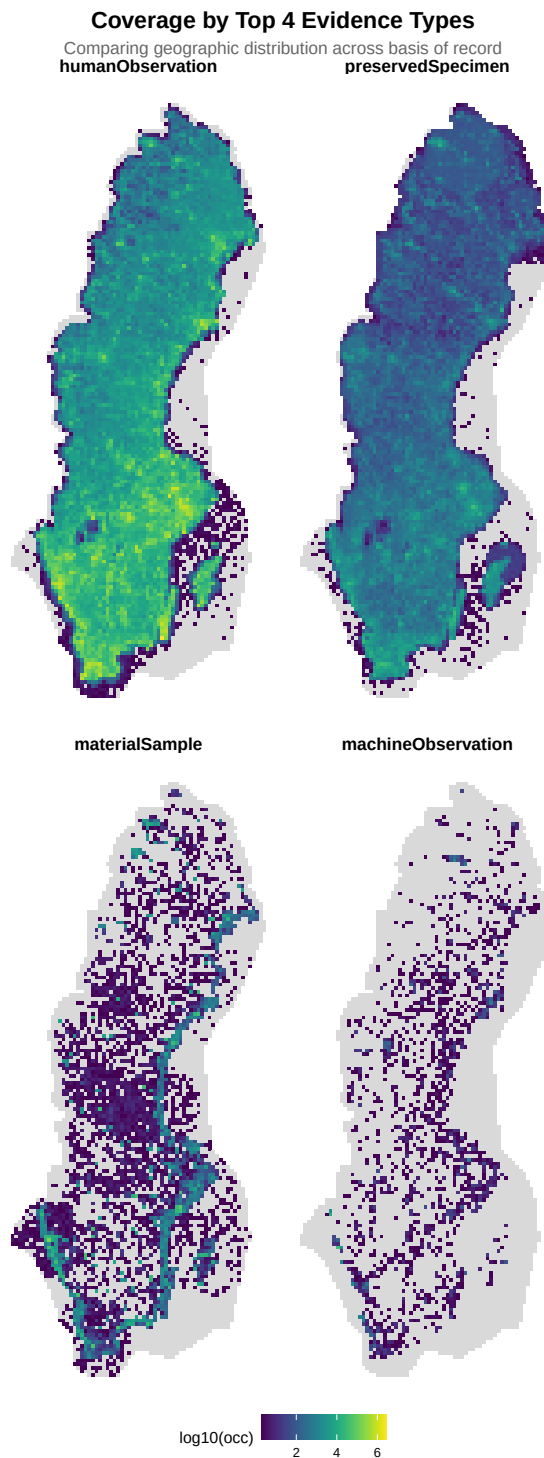
3.1 Zero Coverage Rates

Table 2: Coverage by Basis of Record (10km Grid)

basisofrecord	n_with_data	pct_coverage	n_zero	pct_zero	total_occurrences
humanObservation	5791	82.9	1197	17.1	127827176
preservedSpecimen	5711	81.7	1277	18.3	3782060
materialSample	3391	48.5	3597	51.5	2542895
machineObservation	1451	20.8	5537	79.2	66429
occurrence	1212	17.3	5776	82.7	5447
livingSpecimen	463	6.6	6525	93.4	1841
fossilSpecimen	263	3.8	6725	96.2	7622
materialCitation	253	3.6	6735	96.4	647
observation	38	0.5	6950	99.5	1878



3.2 Coverage Comparison Across Basis Types



4. Priority Areas

4.1 Zero Coverage Cells

708 cells have zero coverage across all basis types.

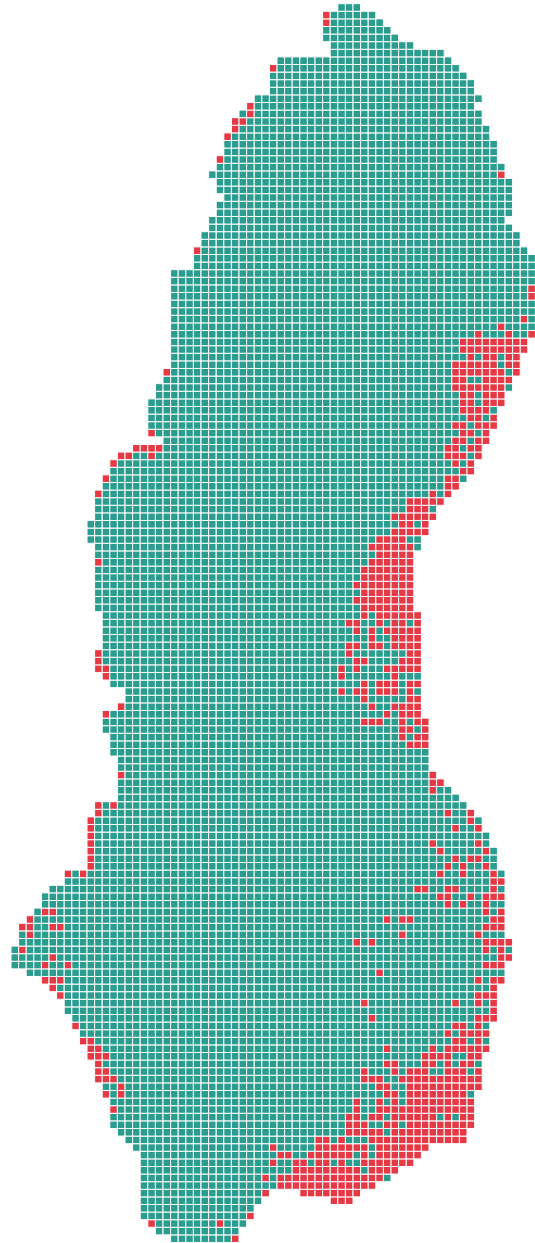
Table 3: Sample of Zero Coverage Cells (Top 20)

grid	eeacellcode	priority_reason	priority_level
grid10km	10kmE431N391	Zero coverage - never sampled	HIGH
grid10km	10kmE431N394	Zero coverage - never sampled	HIGH
grid10km	10kmE432N393	Zero coverage - never sampled	HIGH
grid10km	10kmE432N394	Zero coverage - never sampled	HIGH
grid10km	10kmE432N395	Zero coverage - never sampled	HIGH
grid10km	10kmE434N387	Zero coverage - never sampled	HIGH
grid10km	10kmE434N389	Zero coverage - never sampled	HIGH
grid10km	10kmE434N396	Zero coverage - never sampled	HIGH
grid10km	10kmE435N386	Zero coverage - never sampled	HIGH
grid10km	10kmE435N387	Zero coverage - never sampled	HIGH
grid10km	10kmE435N390	Zero coverage - never sampled	HIGH
grid10km	10kmE435N394	Zero coverage - never sampled	HIGH
grid10km	10kmE435N396	Zero coverage - never sampled	HIGH
grid10km	10kmE436N385	Zero coverage - never sampled	HIGH
grid10km	10kmE436N387	Zero coverage - never sampled	HIGH
grid10km	10kmE436N394	Zero coverage - never sampled	HIGH
grid10km	10kmE437N389	Zero coverage - never sampled	HIGH
grid10km	10kmE437N401	Zero coverage - never sampled	HIGH
grid10km	10kmE438N381	Zero coverage - never sampled	HIGH
grid10km	10kmE439N379	Zero coverage - never sampled	HIGH

4.2 Priority Cells Map

Priority Areas: Zero Coverage Cells

704 cells requiring sampling attention



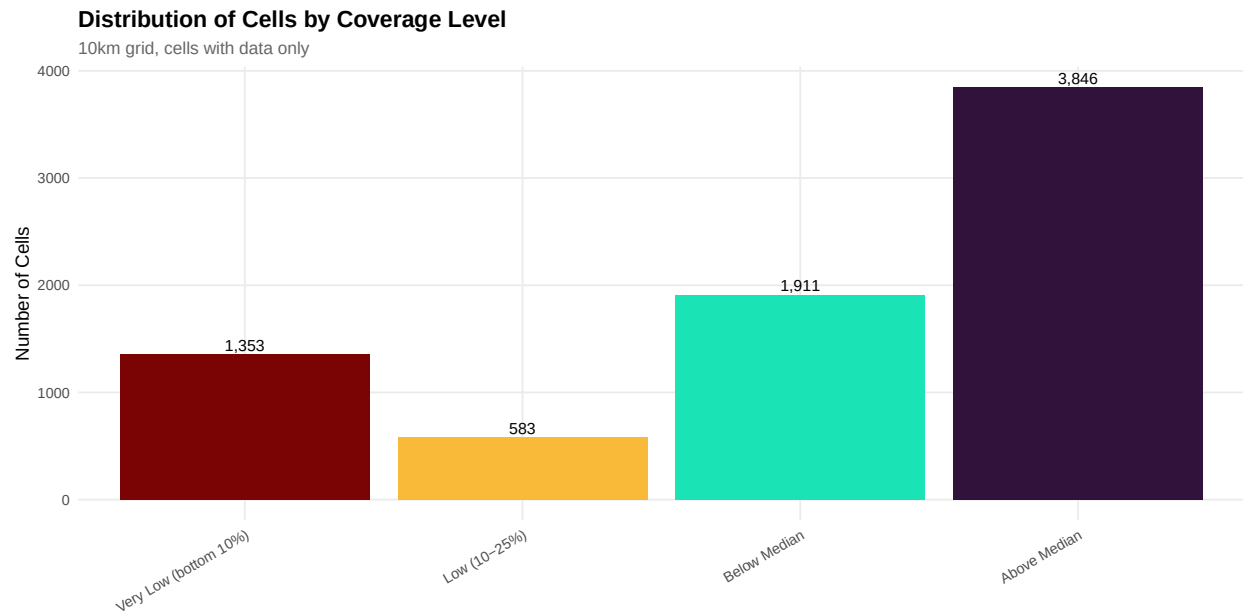
Status ■ Has Data ■ Zero Coverage
Red cells have no occurrence records across any evidence type

4.3 Low Coverage Analysis

Cells with very low coverage (bottom 10%):

##

- Threshold: 1 occurrences- Number of cells: 1,353



5. Key Findings

Spatial Coverage Summary

Grid Coverage:

- grid10km: 89.93% coverage (6,284 / 6,988 cells)
- grid50km: 98.81% coverage (332 / 336 cells)

Priority Areas:

- Zero coverage cells: 708
- Very low coverage cells (bottom 10%): 1,353

Best Coverage by Evidence Type:

- humanObservation: 82.9% coverage
- preservedSpecimen: 81.7% coverage
- materialSample: 48.5% coverage

Recommendations:

1. Target zero-coverage cells for systematic biodiversity surveys
2. Prioritize mountain and northern regions if they show lower coverage
3. Leverage citizen science programs to expand geographic reach

Session Info

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.2
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] targets_1.11.4      httr2_1.2.2          leaflet_2.2.3        plotly_4.12.0        shinyWidgets_0.9.0
## [7] patchwork_1.3.2      DT_0.34.0            tibble_3.3.1          forcats_1.0.1        knitr_1.50
## [13] viridisLite_0.4.2    ggplot2_4.0.1        scales_1.4.0          lubridate_1.9.4      fst_0.9.8
## [19] tidyr_1.3.2          data.table_1.18.0    stringr_1.6.0         readr_2.1.6          dplyr_1.1.4
## [25] cli_3.6.5            glue_1.8.0           purrr_1.2.1          here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] DBI_1.2.3            gridExtra_2.3         rlang_1.1.7           magrittr_2.0.4
## [6] e1071_1.7-17         compiler_4.5.2        systemfonts_1.3.1     callr_3.7.6
## [11] pkgconfig_2.0.3      crayon_1.5.3          fastmap_1.2.0         backports_1.5.0
## [16] labeling_0.4.3       utf8_1.2.6            promises_1.5.0        rmarkdown_2.30
## [21] ps_1.9.1             tinytex_0.58          bit_4.6.0             xfun_0.56
## [26] jsonlite_2.0.0       later_1.4.5           tarchetypes_0.13.2    terra_1.8-93
## [31] prettyunits_1.2.0    R6_2.6.1              bslib_0.9.0           stringi_1.8.7
## [36] jquerylib_0.1.4      Rcpp_1.1.1            leaflet.providers_2.0.0 httpuv_1.6.16
## [41] timechange_0.3.0     tidyselect_1.2.1      rstudioapi_0.18.0     yaml_2.3.10
## [46] curl_7.0.0           processx_3.8.6        withr_3.0.2           S7_0.2.1
## [51] units_1.0-0          proxy_0.4-29          xml2_1.5.2            pillar_1.11.1
## [56] KernSmooth_2.23-26   renv_1.1.6            generics_0.1.4        vroom_1.6.7
## [61] hms_1.1.4            xtable_1.8-4          base64url_1.4         class_7.3-23
## [66] tools_4.5.2          fs_1.6.6              grid_4.5.2            crosstalk_1.2.2
## [71] kableExtra_1.4.0     textshaping_1.0.4    svglite_2.2.2         gt_1.3.0
## [76] sass_0.4.10          digest_0.6.37         classInt_0.4-11       htmlwidgets_1.6.4
## [81] memoise_2.0.1        htmltools_0.5.8.1     lifecycle_1.0.4       httr_1.4.7
## [86] mime_0.13            bit64_4.6.0-1
```